

Contestant Number: \_\_\_\_\_

Time: \_\_\_\_\_

Rank: \_\_\_\_\_



# JAVA PROGRAMMING (340)

## STATE – 2021

### PRODUCTION PORTION:

Stock Portfolio Text Import \_\_\_\_\_ (355 points)

***TOTAL POINTS*** \_\_\_\_\_ (***355 points***)

## Test Time: 90 minutes

You will have ninety (90) minutes to complete your work.

Your name and/or school name should *not* appear on work you submit for grading.

1. Create a folder on the flash drive provided using your contestant number as the name of the folder.
2. Copy your entire solution/project into this folder.
3. Submit your entire solution/project so that the graders may open your project to review the source code.
4. Ensure that the files required to run your program are present and will execute on the flash drive provided.

\*Note that the flash drive letter may *not* be the same when the program is graded as it was when you created the program.

\*It is recommended that you use relative paths rather than absolute paths to ensure that the program will run regardless of the flash drive letter.

The graders will *not* compile or alter your source code to correct for this.  
Submissions that do *not* contain source code will *not* be graded.

**Assumptions to make when taking this assessment:**

- The input file will contain only ASCII characters.
- A test input file will be available and named, "stockText.txt."

**Development Standards:**

- Your Code must use a consistent variable naming convention.
- All methods must be documented with comments explaining the purpose of the method, the input parameters (if any), and the output (if any).
- If you create a class, then you must use Javadoc comments.

## Stock Portfolio Text Import

You are being tasked with creating a software program that can import a text file, read the file, and assign its information to specific variables to construct a list of objects (Stocks) that will be stored in another object, which will serve as the portfolio. The program will print out the entire portfolio once complete. You must use the provided “Stocks.class” and “stocksText.txt” files.

## Portfolio

First, you will create a Portfolio class that creates a list of stocks that are passed to it from the driver class. The Portfolio class will construct Stock objects: you will *not* be required to work with the Stocks class other than using its constructor (provided for you). There are additional methods in this program that are *not* needed. No instance variables are provided for this class.

## Portfolio

Methods and Constructors:

- Portfolio(): this constructor takes no parameters.
- boughtStock(String, String, double, integer): this method will create the Stock objects based upon the values passed to it by the driver class (Stock Driver).
- getPortfolioTotalValue(): this method will return the total value of all the stocks in the list using the stocks getter method getValue() which will return a double value.
- toString(): overwrite this method to access each stock in the list to print out its details. When printing the stocks, utilize their printStock() method (already written for you in another file).
- getPortfolioList(): this method will be called from the driver program (StockDriver) to retrieve the list from the portfolio object.

## Stocks.class

- Stocks(String symbol, String name, double price, integer shares): this constructor takes the parameters listed in this explanation. You will need this for adding stocks to the Portfolio.
- printStock(): this method will print out the stock being called in the portfolio. Use this method in the toString() of the Portfolio.

## StockDriver

Second, you will create a StockDriver class that will import a text file that will transfer the information to a portfolio that will store each record imported as an individual stock object. This class will have the main() method.

## StockDriver

Instance Variables:

- Portfolio port1 – Holds the newly constructed Portfolio object that will store all of the stocks.
- String stockSymbol – Holds three letter symbol of the company.
- String stockName – Holds name of the company.

- double stockPrice – Holds the price of the stock as a non-formatted number.
- integer stockVolume – Holds how many shares of the stock.

In the StockDriver class, you will fill in the main method to import the file, read the file and assign its values to the required instance variables. The information in the text file is organized in the following order: stockSymbol, stockName, stockPrice, stockVolume. You can assume there are no errors in the file. You can write additional helper methods if needed.

- The file will consist of a flag word “STOP”, in any combination of lowercase and uppercase letters, to end the file. For example the flag could be spelled “StOp”, “stop”, “sTOP”, etc.
- Read the file and store the data in the appropriate variables. Pass the variables to the Portfolio object with the method boughtStock(string, string, double, integer).
- After the file is read, call the Portfolio object method getPortfolioList( ) to print the list.

## Input

The input to this problem consists of a text file (stockText.txt). The text file contains a sequence of values to allow the program to create an accurate set of stocks to contain in a list, which is the portfolio. The information in the text file is stored in the following sequence: Stock Symbol, Stock Name, Stock Price, Share Count. The flag word STOP is placed at the end of the text file to terminate the reading from the driver program.

## Input/Output Samples

The output will be created in the following format

*Example input from the text file:*

```
ABC
Google
110.23
25
F
Ford
23.45
100
IBM
International Business Machines
15
5
STOP
```

*Example output to the console (you will NOT be required to create this format...it is already created in the Stock class and can be accessed with the method “public string printStock( )”):*

This is your current portfolio:

```
Symbol: ABC | Company Name: Google | Price: 110.23 | Total Shares: 25 | Total Value: 2755.75
Symbol: F | Company Name: Ford | Price: 23.45 | Total Shares: 100 | Total Value: 2345.0
Symbol: IBM | Company Name: International Business Machines | Price: 15.0 | Total Shares: 5 |
Total Value: 75.0
```

This is the total value of the portfolio: \$5175.75

## Requirements:

1. You will create an application with the main class named StockDriver.
2. You will create a helper class named Portfolio.
3. You must add the class file “Stocks.class” and the text file “stocksText” to your project folder for this application (it will *not* operate correctly without these two files).
4. Your contestant number must appear as a comment at the top of the main source code file.
5. If the input file is *not* found, then the program should display an appropriate message and exit.
6. The program will import and read the text file, storing the results into stocks objects that will be stored in a portfolio object.
7. You must implement methods in Portfolio:
  - a. Constructor that takes no arguments and creates a list that can store Stocks objects.
  - b. boughtStocks( ) method that takes the listed parameters and creates stock objects that are added to the Stocks list.
  - c. getPortfolioTotalValue( ) method that takes no arguments; this method will access each stock in the list and use the stock’s getValue( ) method to get the total value of the portfolio. This method returns a double value.
  - d. toString( ) method that takes no arguments; this method overrides the object based toString method and will access each stock in the list to get its printed format through the stocks printStock( ) method. toString( ) method will return a string that contains the printout for all of the stocks in the list.
  - e. getPortfolioList( ) method prints to the console when called. It will access the toString( ) method inside of Portfolio.
8. You must create StockDriver main( ) method that performs the following:
  - a. Access the text file.
  - b. Read the text file and assign the appropriate value to the correct variables.
  - c. Use the variables to create stocks that are added to the Portfolio object.
  - d. After finished reading the text file: print the entire portfolio by calling its appropriate method to print.
9. If properly executed, the program will display the output like the example above for all of the values in the text file.

**MEMBER ID:**

Your application will be scored on the following criteria:

**Solution and Project**

|                                                     |       |           |
|-----------------------------------------------------|-------|-----------|
| The project is present on the flash drive           | _____ | 10 points |
| The projects main class is named <b>StockDriver</b> | _____ | 10 points |
| The projects helper class is named <b>Portfolio</b> | _____ | 10 points |
| Javadoc comments are used for new classes           | _____ | 10 points |

**Program Execution**

|                                           |       |           |
|-------------------------------------------|-------|-----------|
| The program runs from the USB flash drive | _____ | 15 points |
|-------------------------------------------|-------|-----------|

If the program does *not* execute, then the remaining items in this section receive a score of zero.

|                                                                                                                                  |       |           |
|----------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| The StockDriver class runs and reads the input file                                                                              | _____ | 30 points |
| The StockDriver class displays an error message if the file cannot be found                                                      | _____ | 20 points |
| The StockDriver class stores data fields from stocksText into proper variables                                                   | _____ | 20 points |
| The StockDriver class stops reading the input file when “STOP” is found                                                          | _____ | 20 points |
| Program displays the correct output in the correct order for the records and the field with the formatting from the Stocks class | _____ | 30 points |
| Program displays the correct total value of the portfolio                                                                        | _____ | 20 points |

**Source Code Review**

|                                                                                                                                                                              |       |           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| The source code is properly commented                                                                                                                                        |       |           |
| A comment containing the contestant number is present                                                                                                                        | _____ | 10 points |
| Methods and code sections are commented                                                                                                                                      | _____ | 20 points |
| The StockDriver class creates Portfolio object stores data from stocksText into ArrayList of Stocks with boughtStocks ( ) method                                             | _____ | 30 points |
| Code uses try... catch for exception handling StockDriver class                                                                                                              | _____ | 20 points |
| The Portfolio class constructor creates an ArrayList of Stocks                                                                                                               | _____ | 30 points |
| The Portfolio class method getPortfolioTotalValue( ) calculates and returns the total value of all stocks (it will need to traverse the Stocks ArrayList                     | _____ | 20 points |
| The Portfolio class method toString( ) returns the total printout for all stocks (it will need to traverse the Stocks ArrayList and access each stocks printStock( ) method) | _____ | 20 points |
| Code uses a consistent variable naming convention                                                                                                                            | _____ | 10 points |

**Total Points: 355**